

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

CONTROL SYSTEMS

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) All questions carry marks as indicated against them.
- (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.
- (4) Use graph paper and semilog paper wherever necessary.

1. (a) Using block diagram reduction technique for the system shown in Fig. 1 below, determine the overall transfer function $C(s)/R(s)$.

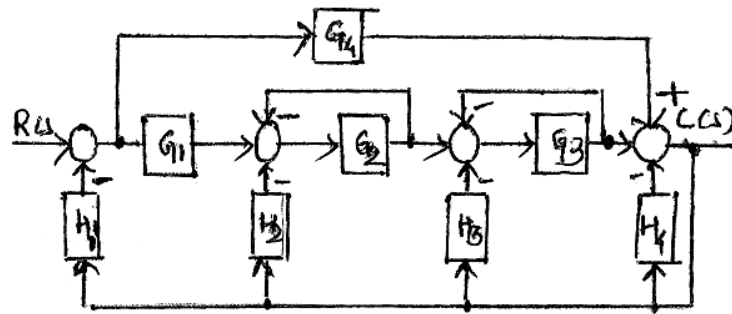


Fig. 1

6(CO2)

- (b) Determine the overall transfer function using Mason's Gain formula for Fig. 2 given below.

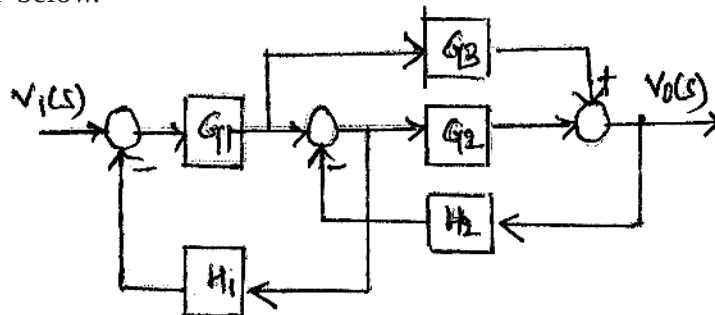


Fig. 2

6(CO2)

2. (a) A second-order feedback system has closed loop poles at $s = -12 \pm j15$. The dc gain of the transfer function is 0.8. Evaluate the following :

- (i) Damping factor.
- (ii) Natural and damped frequencies.
- (iii) Peak overshoot.
- (iv) Peak time.
- (v) Rise time.
- (vi) Settling time.

Also obtain the transfer function.

6(CO2,4)

- (b) A unity feedback system has an open loop gain :

$$G(s) = \frac{K(S + 2)}{S(S^2 + 7S + 42)}$$

Compute the static error constant and the steady state error due to an input of

$$r(t) = t^2 u(t) \quad 6(\text{CO4})$$

3. (a) The closed loop transfer function of a system is :

$$G(S) = \frac{K(S + 1)}{S(S - 1)(S^2 + 4S + 20)}$$

Find the range of K for which the system is stable. Show that the system response can oscillate at two different frequencies, where $H(S) = 1$.

6(CO3)

- (b) Sketch the root locus for a system with open loop transfer function :

$$G(s)H(s) = \frac{K(S + 1)}{S^2(S + 9)} \quad 6(\text{CO3})$$

4. (a) Sketch the polar plot for :

$$G(s) = \frac{10}{(s + 2)(s^2 + 2s + 5)}$$

And find its points of intersection with the real and imaginary axes.

6(CO3,4)

- (b) Sketch Bode plot for a unity negative feedback system ; given by

$$G(s) = \frac{10}{s(s + 1)(s + 10)}$$

6(CO4)

5. (a) For the electrical network shown in Fig. 5(a), obtain state model in the phase-variable form. Select the inductor current as one of the state variables.

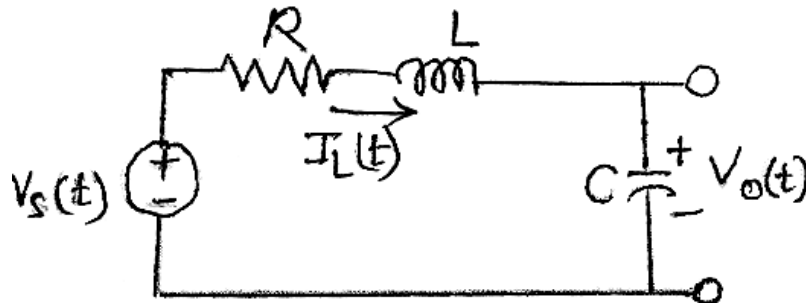


Fig. 5 (a)

6(CO5)

- (b) What do you mean by On-Off linearity with dead zone ? 3(CO1)
- (c) Explain, saturation non-linearity. 3(CO1)

